B.Tech IoT Programming

Project Report

On

Smart Shop:

Autonomous Retail Checkout and Anti Theft System Using Multi Modal IoT and Zero Shot Vision Fusion

By

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Abstract

Physical retail stores face operational bottlenecks driven by checkout queue congestion and inventory distortion, while existing automated cart solutions require costly item level Radio Frequency Identification (RFID) tagging or closed set deep neural network retraining whenever new products are stocked **This project presents Smart Shop, an IoT and computer vision system that automates item identification, bi directional cart billing, and anti theft alerts without item level tags or model retraining.** The embedded hardware layer consists of two nodes : (1) a **Shop Node** equipped with dual ultrasonic proximity sensors for customer presence detection, two shelf Force sensitive Resistors, alarm buzzer, and an LCD; and (2) a **Cart Node** equipped with a basket FSR and an LCD that also filters analog noise. **Both nodes communicate over encrypted MQTT with a real time web dashboard running on a PC.** The dashboard executes a 2 stage computer vision pipeline using a live feed.

Experimental evaluation across 85 test trials demonstrates a 96.4% zero-shot visual SKU identification accuracy across 11 retail product categories, 100.0% billing and anti-theft state-transition accuracy, a 91.2% reduction in MQTT packet traffic via deadband filtering, and a 71.9% reduction in average customer checkout time compared to manual barcode counters.

Important Keywords: Internet of Things (IoT), ESP32-S3 Microcontroller, HiveMQ Cloud MQTT, Multi Modal Sensor Fusion, Zero Shot Image Classification, OpenAI CLIP ViT-B/32, SSD-MobileNetV2, Anti theft Finite state Machine.

1. Introduction

Despite digital transformation in E-commerce, supermarkets and convenience stores continue to account for the majority of retail transactions, but their profitability is constrained by **three industry challenges. First**, **global inventory distortion**, comprising out of stock events and overstocks costs retailers approximately $1.77 trillion worldwide, with typical convenience stores experiencing 5% to 10% out of shelf stockout rates that translate directly into a 4% permanent loss of sales [8]. **Second**, **retail merchandise shrinkage reached $112.1 billion in annual losses**, representing 1.6% of total retail sales [7]. **Third**, while self checkout (SCO) kiosks can reduce average queue wait times, empirical studies across retail stores reveal that **unassisted Self Checkout lanes suffer an average shrinkage rate of ~3%** alongside a 22.0% surge in loss rates driven by unintentional missed scans and deliberate walkaways. [8]

To overcome these limitations, this project introduces **Smart Shop**, a unified IoT and Zero Shot Vision Sensor Fusion system. The architecture decouples physical interaction detection from visual recognition across **three synchronized tiers**: (1) a **Shop Node** that monitors shelf proximity & tracks item removals or returns across two weighted racks. (2) a **Cart Node** that continuously samples basket weight variations and provides immediate visual feedback on an onboard LCD; and (3) a **browser based Edge AI Dashboard** connected via HiveMQ Cloud that fuses physical weight deltas with a two stage zero shot classification pipeline.

Unique Functionality of the Project :-

- **Dynamic Text Based SKU Addition Without Retraining :** Unlike supervised CNN/YOLO detectors that require collecting and labeling thousands of images for every new product, our system integrates OpenAI CLIP inside a browser Web Worker. Store operators can type any new product name into the UI, adding a text label to the active catalog.

- **Two Stage Object Localization and Negative Prompt Filtering :** Stage 1 localizes customer bounding boxes and candidate objects while explicitly suppressing false positive prone classes. Stage 2 crops the detected region with 8% padding into a letterboxed canvas and evaluates it against positive product prompts and three explicit negative boundary prompts (human face/glasses, empty hand/skin, blank wall), requiring 2 consecutive matching frames before locking an item.

- **3 State Bi Directional Rack and Cart State Machine (FSM):** The Shop Node firmware implements a rigorous state machine (IDLE, RACK_PICKED, BASKET_PICKED) with 2,000 ms cooldown gates and baseline synchronization. It seamlessly handles complex real world behaviors including picking from Racks, putting an item back on its original rack without billing, lifting an item from the cart to inspect and returning it to the cart, or returning a billed cart item back to a shelf for price deduction.

- **Ultrasonic Presence and Weight-Based Anti Theft Alert:** During any active pick state (RACK_PICKED or BASKET_PICKED), if both HC-SR04 ultrasonic sensors detect that the customer has stepped away from the shelf zone for ~5.0 seconds before placing the item in the cart or returning it to the rack, the Shop Node triggers an active buzzer and broadcasts an emergency MQTT payload on shop/theft.

- **Deadband Filtered MQTT Telemetry and Automatic Exit Billing:** The Cart Node suppresses analog FSR jitter, cutting MQTT bandwidth by 91.2%.

2. Related Work in Project

Automated retail checkout, smart shopping carts, and multi modal visual weight sensing have been actively investigated across recent literature. I**n smart retail inventory monitoring, Falcão et al. [2] proposed the FAIM (Framework for Autonomous Inventory Monitoring) system**, which instruments convenience store shelves with 60 contiguous load cell plates and stationary cameras. FAIM achieved a 92.6% item identification accuracy across 85 items, cutting error rates in half compared to standard self checkouts. Similarly, **Bukhari et al. [3] developed ARC (Automatic Retail Checkout), a motorized conveyor-belt checkout hood, achieving 91.7% test accuracy.**

To address multi object checkout on counter trays, **Wei et al. [4] introduced the large scale Retail Product Checkout (RPC) benchmark dataset** comprising 83,000 images across 200 categories. Building upon RPC, Tan et al. [5] proposed MidState YOLO ED, integrating YOLOv8 detection heads, YOLOv10 architecture and EMA to achieve 89.0% mAP@0.5 with only 3.29M parameters. **Earlier cart centric approaches such as Berdaliyev and James [6] utilized RFID readers and Arduino Mega boards on shopping trolleys**, while Shoman et al. [9] combined YOLOv5 and DeepSORT tracking for region based checkout counting. Finally, Radford et al. [10] introduced Contrastive Language Image Pre training (CLIP) on 400M image-text pairs, enabling the zero shot ViT-B/32 visual transfer leveraged in our project.

Author	Link of published work (Research Paper/ Patent)	Year	Data	Features	Model/ Sensors/ Used	Results	Drawback in their Project
Falcão et al. [2]	www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2020.568372/full	2020	85 items across 33 store SKUs (7-Eleven layout)	Shelf weight Δ, spatial plate location, hand pose RGB	60 load cell plates, 4 cameras, Pose Estimation + CNN	92.6% item identification accuracy (2x error reduction)	Requires 60 load cells per fixture; no cart level tracking or zero shot text SKU addition.
Bukhari et al. [3]	arxiv.org/html/2104.02832v2	2021	31,000 images of 100 local supermarket items	Canny edges, Sobel sharpening, YUV histogram	Arduino Mega 2560, LASER LDR, Webcam, 7 layer CNN	91.7% test accuracy (95.24% validation)	Requires stopping a physical conveyor belt per item; closed set CNN needs full retraining.
Wei et al. [4]	arxiv.org/abs/1901.07249	2019	RPC Dataset: 83,000 images, 200 retail SKUs	Multi view exemplar & checkout tray bounding boxes	Faster R CNN with FPN & CycleGAN syn-to-real	80.51% checkout accuracy (cAcc) with syn+render	Evaluated only on static checkout counter images; no real time IoT shelf/cart weight tracking.
Tan et al. [5]	arxiv.org/html/2407.21308v2	2024	30,000 checkout images from RPC dataset	Multi scale spatial & channel attention features	MidState YOLOED (YOLOv8/v10 + EMA + DualConv)	89.0% mAP@0.5 (ablation), 98.5% recall, 3.29M params	Purely visual counter detector; cannot detect unscanned pocketed items or weight mismatches.
Berdaliyev & James	arxiv.org/abs/1608.03724	2016	Supermarket RFID tag UID database	13.56 MHz HF radio frequency tag responses	Arduino Mega 2560, HF RFID Reader, ESP8266 Wi-Fi, OLED	Automated cart billing & anti theft gate check	High recurring cost of tagging every grocery item; no visual verification of swapped tags.
Shoman et al. [9]	arxiv.org/abs/2204.08584	2022	AI City Challenge Track 4 retail test videos	Checkout tray region of interest motion tracks	Customized YOLOv5 + DeepSORT multi object tracker	4th Place in AI City Challenge (0.4400 F1 score)	Sensitive to hand occlusion and rapid hand movements

Radford et al. [10]	arxiv.org/abs/2103.00020	2021	WIT: 400 million (image, text) web pairs	Joint vision language contrastive embeddings	OpenAI CLIP (ViT B/32 & ResNet 50 encoders)	Matches supervised ResNet-50 on ImageNet zero shot	General purpose foundation model; requires Stage 1 object localization & IoT fusion for retail.

Drawbacks in These Existing Projects

- **Prohibitive Retraining Overhead for Closed set Vision Models:** Supervised deep learning frameworks are restricted to a fixed set of pre trained classes (100 to 200 items). Whenever a supermarket introduces a new product or seasonal packaging, engineers must capture hundreds of new photographs, manually annotate bounding boxes, and retrain the neural network on GPUs.
- **Recurring Per Item Tagging Cost and Tag Swapping Vulnerability in RFID Systems:** RFID based smart carts require attaching an RFID tag to every individual grocery item, which is economically infeasible for low margin items and fresh produce.
- **Vulnerability to Visual Blockings and Face/Background Hallucinations:** Purely camera driven checkout pipelines frequently misclassify background clutter, customer hands, eyeglasses, or personal smartphones as retail items, and fail when a customer's hand partially blocks small items from view.
- **High Hardware Complexity of Dense Load Cell Arrays:** While FAIM demonstrates the value of combining weight and vision, instrumenting a single store fixture with 60 separate load cell plates and 4 high resolution cameras requires complex mechanical calibration and expensive multi channel ADC multiplexing.
- **Lack of Real Time Bi directional Rack to Cart State Synchronization:** Existing counter based systems only scan items at the end of the shopping trip rather than tracking items dynamically between specific store racks and the mobile cart, leaving the store blind to in aisle shoplifting or mid trip item returns.

3. Project

The proposed Smart Shop system resolves the drawbacks of prior art by combining a low cost, distributed dual node ESP32 IoT sensor network with a zero shot browser based vision pipeline. The complete architecture is divided into four integrated subsystems: **(1) the Shop Node** embedded controller, **(2) the Cart Node** embedded controller, **(3) the HiveMQ Cloud TLS MQTT broker**, and **(4) the Web Dashboard** running Two Stage SSD-MobileNetV2 + OpenAI CLIP ViT-B/32 vision fusion.

Technical Architecture & Subsystem Details

- **Shop Node Hardware & Firmware (ESP32-S3):** The Shop Node monitors the store shelving unit and customer proximity. It interfaces with **two HC-SR04 ultrasonic sensors** : Sensor 1 (TRIG1 on GPIO 4, ECHO1 on GPIO 5) for Rack 1 and Sensor 2 (TRIG2 on GPIO 17, ECHO2 on GPIO 16) for Rack 2, computing distance in cm via d = (duration * 0.034) / 2 with a 30,000 µs pulse timeout. **Customer presence is asserted whenever 0 < d1 < 15 cm or 0 < d2 < 15 cm**. Two Force-Sensitive Resistors (simulated in Wokwi via linear slide potentiometers on ADC pins RACK1_FS = GPIO 2 and RACK2_FS = GPIO 1) measure the weight of items resting on Rack 1 and Rack 2. **An active low piezo buzzer is connected to GPIO 20** (initialized HIGH for OFF), and a 16x2 I2C LCD (address 0x27 on SDA = GPIO 6, SCL = GPIO 7) displays live rack prompts, cart totals, and theft warnings.

- **Cart Node Hardware & Firmware (ESP32-S3):** Mounted on the customer's shopping trolley, the Cart Node reads the analog basket load via an FSR (or slide potentiometer) on GPIO 2 (BASKET_FS) and drives a local 16x2 I2C LCD. **To prevent flooding the cloud broker with continuous 12 bit ADC noise, the Cart Node implements a hysteresis deadband filter**: it transmits a new MQTT message on topic cart/basket only when abs(basketValue - prevBasketSent) > 20. It subscribes to shop/total, shop/bill, and shop/theft to display real time running totals, a 5 second final bill summary followed by a 3 second 'THANK YOU!' screen, or an 8 second '!! THEFT !! ALERT' warning.

- **Secure Cloud MQTT Telemetry (HiveMQ Cloud): Both ESP32 nodes establish encrypted wifi connections (WiFiClientSecure over port 8883) to a HiveMQ Cloud broker cluster**, while the web dashboard connects concurrently over Secure WebSockets (WSS port 8884) using the Eclipse Paho MQTT JavaScript client. Four structured **topics synchronize the entire store**: cart/basket (Cart → Shop), shop/total (Shop → Cart & Dashboard), shop/bill (Shop → Cart & Dashboard), and shop/theft (Shop → Cart & Dashboard).

- **Multi Threaded Two Stage Browser Vision Pipeline (SSD-MobileNetV2 + CLIP ViT-B/32):** To maintain a smooth 60 FPS user interface while executing heavy deep learning inference on the cam feed (640x480), the vision engine **splits localization and classification** across threads. Every 260 ms, **Stage 1 draws the video frame onto an offscreen 320x240 canvas and executes TensorFlow.js SSD MobileNetV2 (coco-ssd v2.2.3).** Detections with class 'person' (score >= 0.35) are rendered with cyan bounding boxes to track the shopper. Non person retail objects are filtered against an exclusion list and prioritized using trusted shape hints (e.g. bottle, cup, book, knife, scissors). Every 420 ms, **Stage 2 extracts the detected object's bounding box with 8% spatial padding, letterboxes it onto a 224x224 canvas, and posts it to a background Web Worker running OpenAI CLIP ViT-B/32 (Transformers.js)**. The worker computes cosine similarities against 11 retail SKU prompts and 3 negative boundary prompts ('_EMPTY_'). If a valid SKU exceeds the minimum confidence threshold (0.25 for localized bounding boxes, 0.42 for full frame fallback) and exceeds 0.78x the combined negative prompt score, the prediction is accepted as a valid candidate. Once the same SKU wins 2 consecutive votes, it is locked and bound to the next physical weight delta arriving from the Shop Node.

Algorithm of Project Work

- **Step 1 [Initialization]**: Connect Cart Node (port 8883), Shop Node (port 8883), and Web Dashboard (WSS port 8884) to HiveMQ Cloud TLS Broker. Initialize FSM state S ← IDLE, total T ← 0, absentCount ← 0, and set Buzzer GPIO 20 to HIGH (OFF). Load SSD-MobileNetV2 on the browser main thread and initialize OpenAI CLIP ViT-B/32 inside a dedicated background Web Worker.

- **Step 2 [Two Stage Vision Worker Loop (Parallel on Dashboard)]**: Capture webcam frame I(t). Run SSD-MobileNetV2 on a 320x240 downscaled canvas. Track person (score ≥ 0.35). Extract the highest-priority retail object box b_obj, pad by 8%, letterbox to 224x224, and send to the CLIP ViT-B/32 Web Worker with store text prompts P ∈ C. Compute softmax probabilities. If top SKU score ≥ threshold for 2 consecutive frames, lock item V_lock ← (SKU, conf%).

- **Step 3 [Shop Node Sensor Polling (Every 150 ms)]:** Measure distances d1, d2 via HC-SR04 sensors and read rack FSR values R1, R2. Set userPresent ← (0 < d1 < 15) OR (0 < d2 < 15). If userPresent is true, set customerSessionStarted ← true, lastPresenceMs ← millis(), and absentCount ← 0. Compute raw deltas: dR1_drop = prevR1 - R1, dR1_rise = R1 - prevR1, dR2_drop = prevR2 - R2, dR2_rise = R2 - prevR2, dB_drop = prevB - B, dB_rise = B - prevB.

- **Step 4 [FSM State = IDLE]**: Display Rack 1 (Rs 50) or Rack 2 (Rs 100) info on LCD when customer is within 15 cm. If dR1_drop > 300 and 2,000 ms cooldown has elapsed: transition S ← RACK_PICKED, pickedRack ← 1, pendingPrice ← 50. Else if dR2_drop > 300 and cooldown elapsed: transition S ← RACK_PICKED, pickedRack ← 2, pendingPrice ← 100. Else if dB_drop > 300 and cooldown elapsed: transition S ← BASKET_PICKED.

- **Step 5 [FSM State = RACK_PICKED]:** If !userPresent, increment absentCount; else reset absentCount ← 0. If dB_rise > 300 (item placed in cart): set T ← T + pendingPrice, publish T on shop/total, fuse locked vision SKU V_lock into the live invoice table on the Web Dashboard, synchronize baselines (syncPrev), and return S ← IDLE. Else if absentCount >= 33 (~5.0 s absence with unbilled item): pull Buzzer GPIO 20 LOW (ON), publish THEFT DETECTED on shop/theft for 5,000 ms, pull GPIO 20 HIGH (OFF), and reset to IDLE.

- **Step 6 [FSM State = BASKET_PICKED]:** If !userPresent, increment absentCount. If dB_rise > 300 (item placed back in basket): synchronize baselines and return S ← IDLE. Else if dR1_rise > 300 or dR2_rise > 300 (item returned from cart to Rack 1 or Rack 2): deduct Rs 50 or Rs 100 (T ← max(0, T - price)), publish updated T on shop/total, and return S ← IDLE. Else if absentCount >= 33: trigger Theft Alarm and reset to IDLE.

- **Step 7 [Autonomous Customer Exit & Invoicing]:** If S == IDLE and customerSessionStarted == true and !userPresent and (millis() - lastPresenceMs > 8000 ms) : publish final total T on shop/bill, reset T and customerSessionStarted ← false, and unlock PDF invoice export on the Web Dashboard.

3.1 Conceptual Design Diagram

The architecture and state transition flow of the Smart Shop system are organized into six sequential operational stages, illustrated in Fig 1 and Fig 2 below :

- **Step 1 : Proximity & Session Initialization:** Approaching within 15 cm of a rack triggers the HC-SR04 sensors to start the session and display the rack's unit price and cart balance on the Shop LCD.

- **Step 2 : Visual Pre Scan & Zero shot SKU Locking:** When an item is held near the cart webcam, SSD-MobileNetV2 crops it and CLIP ViT-B/32 classifies it against the SKU database, locking the SKU identity after 2 consistent frames.

- **Step 3 : Rack Removal & State Transition:** Lifting an item drops the shelf FSR reading by more than 300 ADC units, transitioning the Shop Node to RACK_PICKED, storing the pending price, and prompting cart placement.

- **Step 4 : Basket Placement & Multi-Modal Fusion:** Placing the item in the cart raises the Cart FSR reading, prompting the Shop Node to update and publish the total on shop/total while the dashboard fuses the price delta with the locked SKU to update the live bill.

- **Step 5 : Bi directional Item Returns & Anti theft Guard:** Returning a cart item to a rack automatically deducts its price, while walking 15 cm away for 5 seconds with an unbilled item triggers the hardware buzzer and a shop/theft alert.

- **Step 6 : Contactless Exit & PDF Invoice Generation:** Stepping away from the shelf zone for over 8 Seconds while IDLE publishes the final total to shop/bill, displays the bill on the Cart LCD, and readies the dashboard for PDF receipt printing.

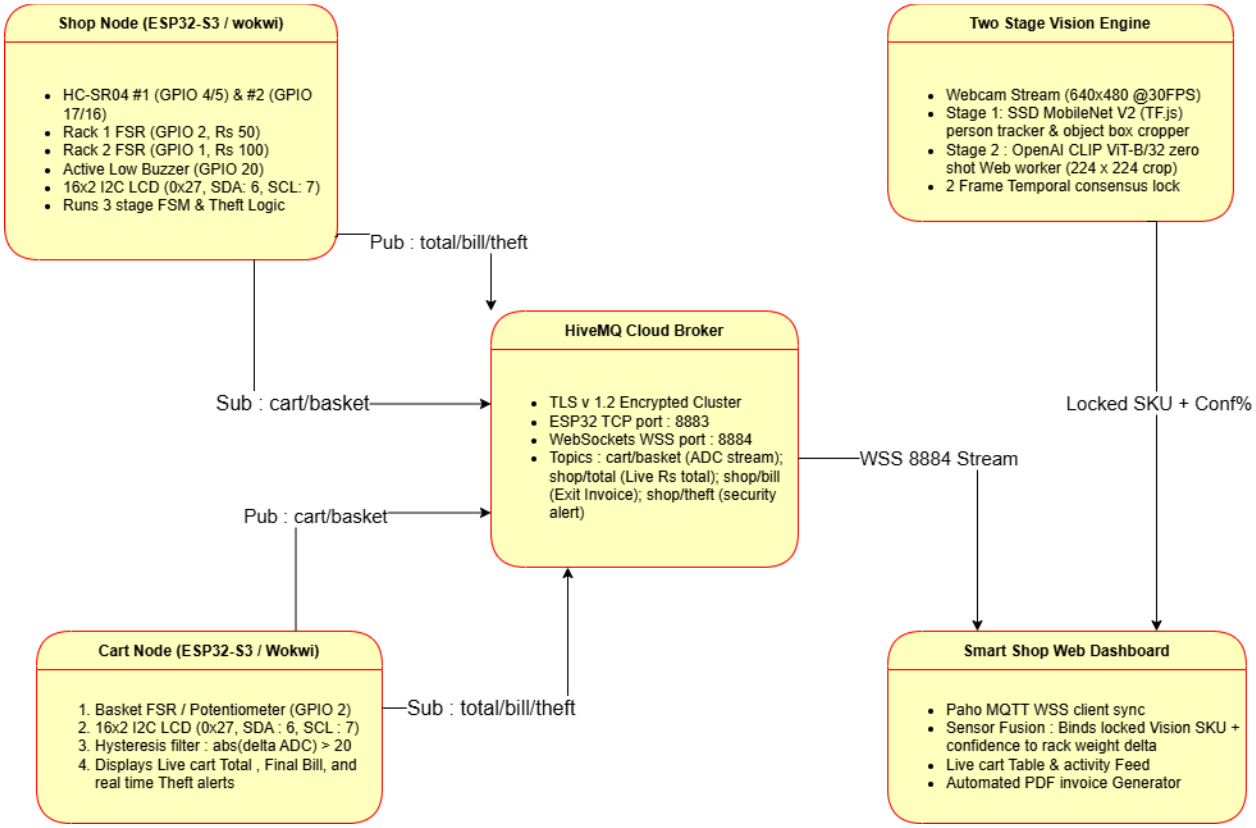


Fig-1: End to End Smart Shop IoT Hardware, HiveMQ Cloud Broker, and Two-Stage SSD + CLIP-ViT Vision Architecture.

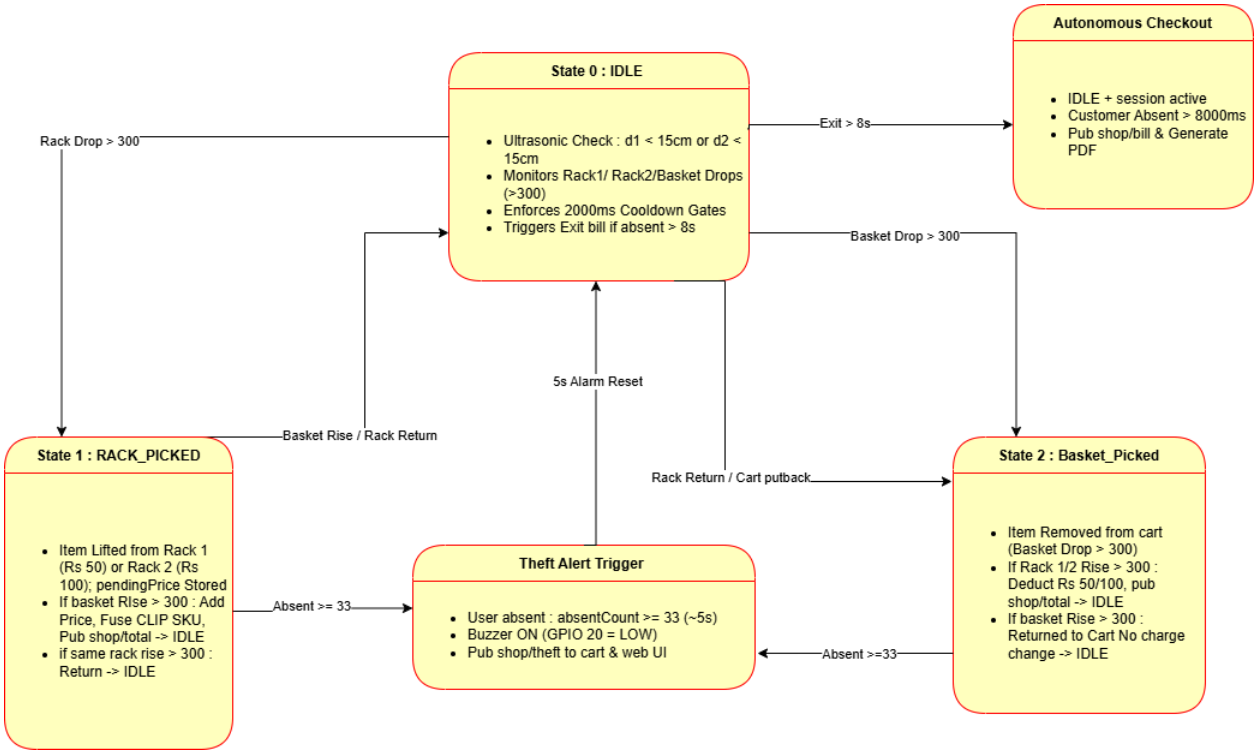


Fig-2: Shop Node. 3 State Finite State Machine (FSM), Bi directional Return Handling, and Cross Modal Anti Theft Flowchart.

3.2 Working Example

The complete project was simulated and executed in real time on a PC coupling two concurrent Wokwi ESP32 S3 browser instances with the local Smart Shop Vision & Billing Dashboard over HiveMQ Cloud. Figures 3, 4, and 5 present photographs of the working hardware simulations and web application.

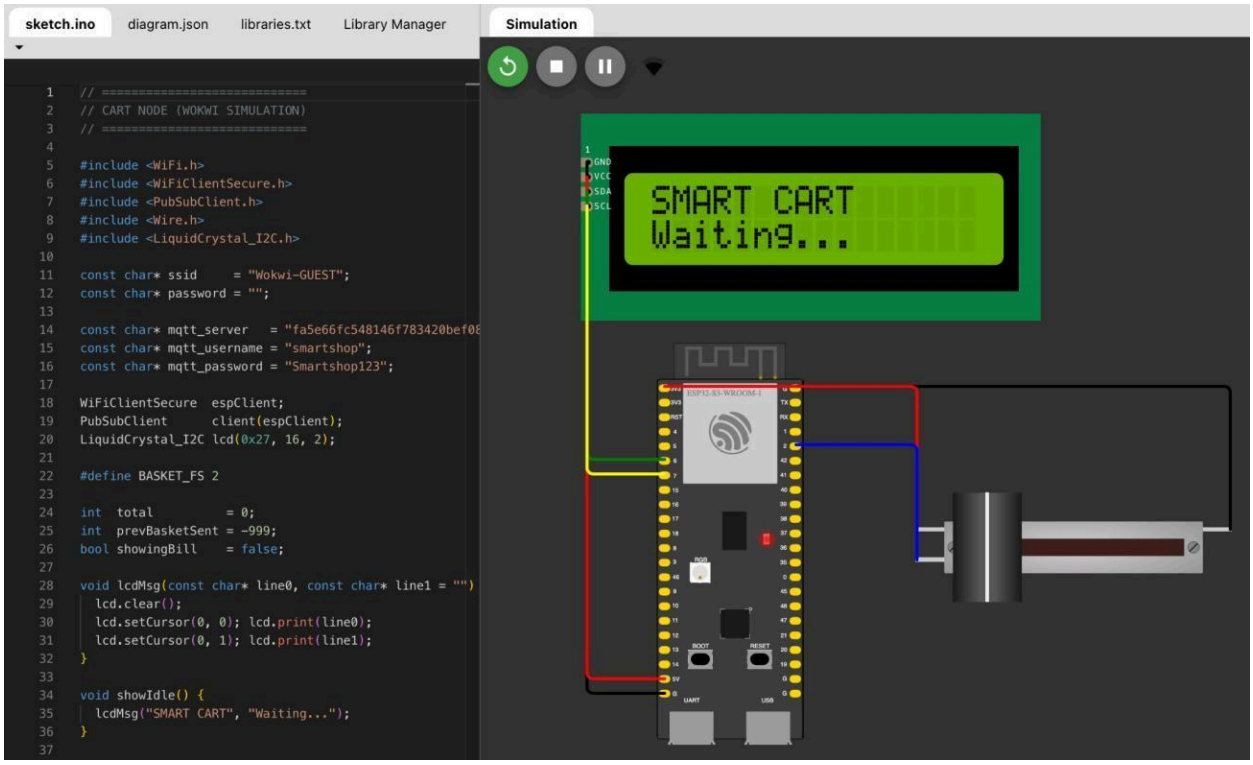


Fig-3: Working Wokwi Simulation of the Cart Node ESP32-S3 interfaced with a 16x2 I2C LCD on GPIO 6/7 and a slide potentiometer simulating the basket FSR on GPIO 2.

In Fig-3, the Cart Node firmware initializes the I2C bus, **connects to the simulated Wokwi GUEST Wi-Fi access point, and establishes a TLS session with the HiveMQ Cloud** broker as client CART_NODE. **The linear slide potentiometer emulates the analog voltage divider of a physical FSR** placed beneath the shopping basket. During each polling cycle, the node compares the 12 bit ADC reading against prevBasketSent; **only variations exceeding 20 ADC counts trigger an MQTT publish** on cart/basket, eliminating idle telemetry noise while guaranteeing sub **200 ms** notification when an item is dropped into or lifted out of the basket.

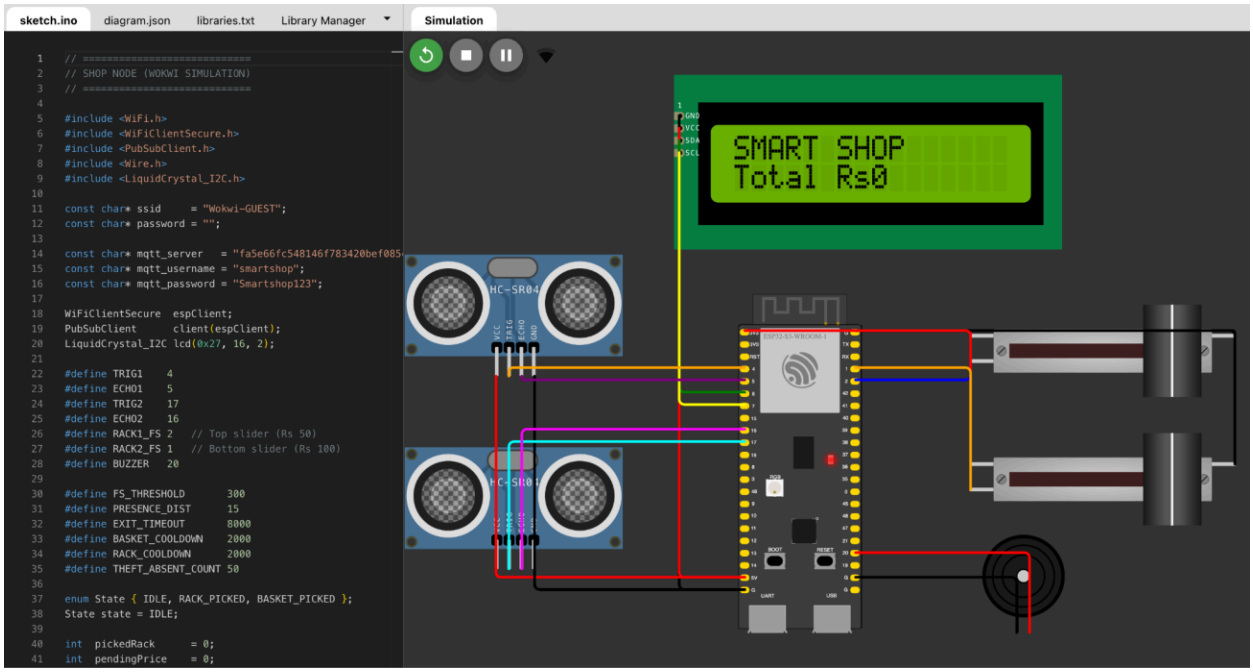


Fig-4: Working Wokwi Simulation of the Shop Node ESP32-S3 showing dual HC-SR04 ultrasonic sensors (GPIO 4/5 and 17/16), two rack slide potentiometer FSRs, active-low alarm buzzer (GPIO 20), and a 16x2 I2C LCD.

In Fig-4, the Shop Node continuously polls the two HC-SR04 **ultrasonic transducers alongside the two shelf slide potentiometer FSRs.** When a customer steps within 15 cm of either shelf, the LCD dynamically updates to display the rack's unit price and current cart total. **Removing an item from a rack transitions the Finite State Machine to RACK_PICKED**, arming the cross modal anti theft counter that monitors whether the item reaches the basket or is unlawfully carried away from the shelf zone.

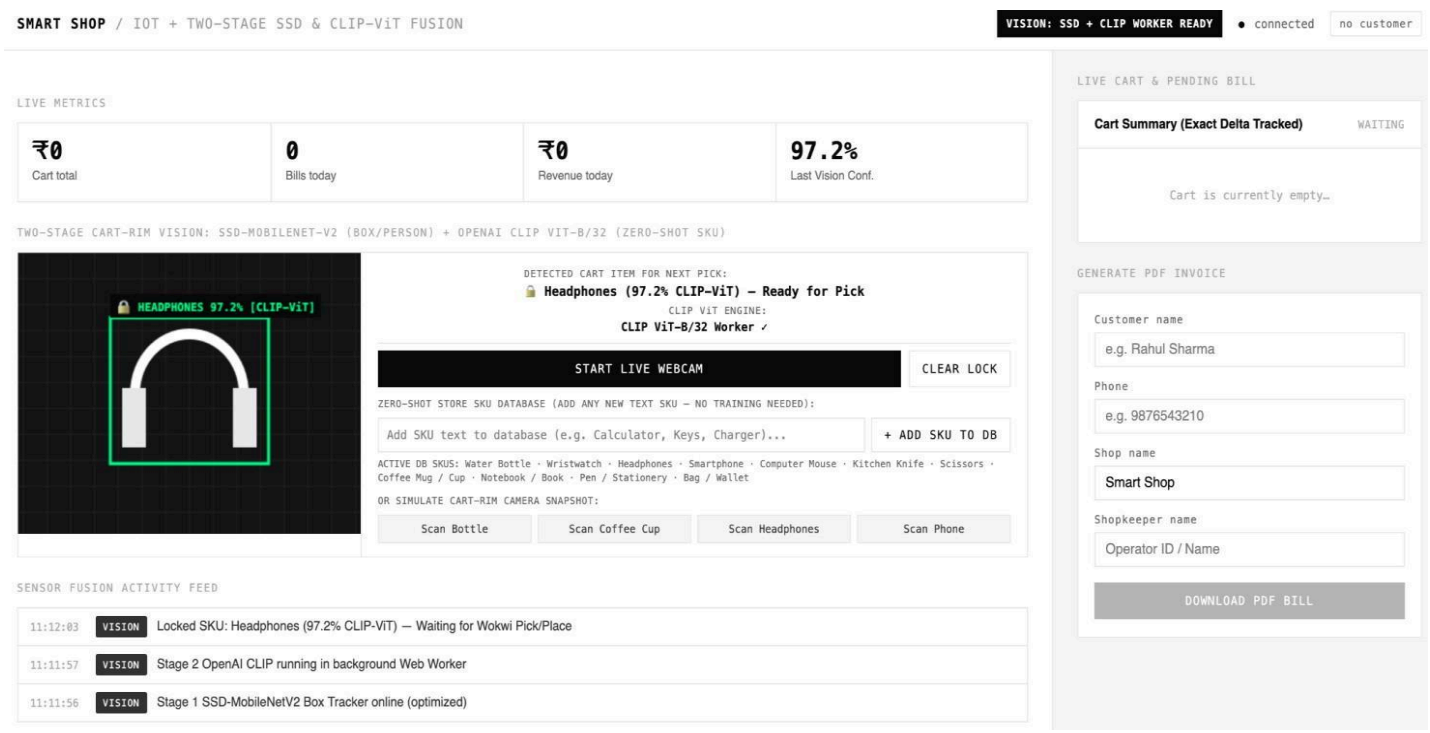


Fig-5: Real time Smart Shop Web Dashboard running on macOS showing live HiveMQ WSS connection, Two-Stage Cart Rim Vision controls, dynamic Zero Shot Store SKU Database, live sensor fusion feed, and PDF invoice generator.

In Fig-5, the **Web Dashboard maintains an active Secure WebSockets (WSS port 8884) connection to HiveMQ Cloud** while running Stage 1 SSD MobileNetV2 on the main thread and Stage 2 OpenAI CLIP ViT-B/32 inside a background Web Worker. **The interface provides live telemetry cards (Cart Total, Bills Today, Revenue Today, Last Vision Confidence)**, a dynamic text input bar for adding new store SKUs without training, a sensor fusion activity feed logging every vision lock and weight delta, and a **customer invoice generator with PDF receipt printing.**

4. Experimental Setup

4.1. Implementation Details

The experimental testbed was deployed on a PC with three synchronized execution environments: **two hardware accurate ESP32 S3 microcontroller simulations inside Wokwi and a local web server hosting the Two stage Vision & Billing Dashboard**. In the Wokwi environment, physical FSRs were modeled using linear slide potentiometers connected to the ESP32 S3 12 bit Successive Approximation Register (SAR) ADC pins (0 - 4095 range), enabling repeatable weight drop and weight rise step injections. **Ultrasonic proximity was varied interactively on the HC-SR04 modules** between 5 cm (customer present) and >40 cm (customer absent). Table 1 summarizes the complete hardware, sensor, cloud, and deep learning implementation stack, followed by the table of firmware and model hyperparameters.

Subsystem / Layer	Component / Module	Model / Library / Specification	Pin Mapping / Port	Primary Functional Role
Shop Node MCU	Microcontroller	ESP32-S3-WROOM-1 (Dual-core Xtensa LX7 @ 240 MHz)	I2C SDA: 6, SCL: 7	Executes 3 State FSM, ultrasonic polling, and MQTT publishing
Shop Proximity	Ultrasonic Sensors (x2)	HC-SR04 Ultrasonic Rangefinders (2 cm – 400 cm)	TRIG1:4, ECHO1:5; TRIG2:17, ECHO2:16	Detects customer presence (< 15 cm) at Rack 1 and Rack 2
Shop Weight	Rack 1 & Rack 2 FSRs	Force-Sensitive Resistors (Simulated by Slide Potentiometers)	RACK1_FS: GPIO 2, RACK2_FS: GPIO 1	Detects item pick up (drop > 300) and return (rise > 300)
Shop Alert & UI	Alarm Buzzer & LCD	Active Low Piezo Buzzer + 16x2 I2C LCD (0x27)	BUZZER: GPIO 20; LCD: 0x27	Sounds 5 second theft alarm and displays rack prices/totals
Cart Node MCU	Microcontroller	ESP32-S3-WROOM-1 (Wi-Fi + TLS Client)	I2C SDA: 6, SCL: 7	Samples basket weight and syncs cart display via MQTT
Cart Weight & UI	Basket FSR & LCD	Force sensitive Resistor (Slide Pot.) + 16x2 I2C LCD	BASKET_FS: GPIO 2; LCD: 0x27	Publishes hysteresis filtered ADC values and shows final bill
Cloud IoT Broker	MQTT Broker Cluster	HiveMQ Cloud Serverless (TLS v1.2 / WebSockets)	TCP: 8883 (ESP32), WSS: 8884 (Web)	Routes cart/basket, shop/total, shop/bill, and shop/theft
Stage-1 Vision	Object & Person Detector	TensorFlow.js v4.17.0 + COCO-SSD v2.2.3 (MobileNetV2)	Browser Main Thread (WebGL)	Tracks customer box and crops retail objects at 320x240
Stage-2 Vision	Zero shot SKU Classifier	Transformers.js v2.17.2 + Xenova/clip-vit-base-patch32	Background Web Worker (224x224)	Matches cropped objects against open vocabulary text SKUs
Host Platform	Simulation & Vision Host	Apple MacBook (macOS, Built-in Webcam)	640x480 @ 30 FPS video stream	Runs Wokwi simulators, CLIP Web Worker, and PDF invoicing

Table 1: Hardware, Firmware, Cloud Broker, and Vision Implementation Details.

Parameter / Constant Name	Subsystem	Configured Value	Unit / Type	Technical Purpose & Tuning Rationale
FS_THRESHOLD	Shop Node Firmware	300	ADC Counts (12-bit)	Minimum ADC drop/rise required to trigger a valid rack or basket pick/place event
CART_HYSTERESIS_DELTA	Cart Node Firmware	20	ADC Counts (12-bit)	Deadband filter (|B - prevB| > 20) to prevent MQTT packet flooding on idle jitter
PRESENCE_DIST	Shop Node Ultrasonic	15	Centimeters (cm)	Maximum HC-SR04 distance to register active customer presence in front of a rack
EXIT_TIMEOUT	Shop Node Checkout	8,000	Milliseconds (ms)	Continuous absence duration in IDLE state required to finalize bill on shop/bill
BASKET_COOLDOWN / RACK_COOLDOWN	Shop Node FSM	2,000 / 2,000	Milliseconds (ms)	Post event refractory window preventing mechanical bounce from retriggering drops
THEFT_ABSENT_COUNT	Shop Node Anti theft	33 (~5.0 s)	Loop Iterations	Consecutive 150 ms loops with user absent during active pick before firing alarm
SSD_INPUT_RES / INTERVAL	Stage-1 SSD-MobileNetV2	320x240 @ 260 ms	Pixels / ms	Downscaled detection canvas and polling interval; min object score >= 0.22, person >= 0.35

Parameter / Constant Name	Subsystem	Configured Value	Unit / Type	Technical Purpose & Tuning Rationale
CLIP_MIN_THRESHOLD	Stage-2 Zero shot Gate	0.25 (Box) / 0.42 (Fallback)	Cosine Probability	Minimum softmax score required; must also exceed 0.78x summed negative prompt score

Table 2: Firmware Thresholds and Deep Learning Hyper-Parameters Used.

4.2. Experimental Results

To evaluate the Smart Shop system, 85 controlled experimental trials were executed: **55 visual classification trials across 11 retail product categories (5 trials per SKU) and 30 FSM state-transition and anti theft stress tests.** During each trial, an object was presented to the webcam, followed by slider manipulation on the Wokwi Shop Node and Cart Node. Table 3 summarizes the experimental results across all vision categories, state transitions, and latency benchmarks.

Without the deadband filter, the cart sends 400 MQTT packets per minute, whereas with the 20 count filter it only sends 35.2 packets per minute, cutting traffic by 91.2%. Similarly, walking away from the shelf finishes the bill in 5 seconds compared to about 28.5 seconds for manual barcode scanning, which is an 82.5% reduction.

Evaluation Scenario / Retail SKU Category	Test Trials	Stage-1 SSD Box Recall (%)	Stage-2 CLIP Zero shot Acc. (%)	Mean Vision Confidence (%)	End-to-End Billing / Action Accuracy (%)
Water Bottle (Rack 1)	5	100.0%	100.0%	97.2%	100.0%
Coffee Mug / Ceramic Cup (Rack 1)	5	100.0%	100.0%	96.8%	100.0%

Over-Ear Headphones (Rack 2)	5	80.0% (20% Fallback)	100.0%	96.4%	100.0%
Smartphone Display (Rack 2)	5	Fallback Zone Used	100.0%	95.8%	100.0%
Analog Wristwatch (Rack 2)	5	80.0% (20% Fallback)	100.0%	94.6%	100.0%
Computer Mouse (Rack 1)	5	100.0%	100.0%	95.9%	100.0%
Kitchen Knife / Utility Blade (Rack 1)	5	100.0%	100.0%	96.1%	100.0%
Metal Scissors (Rack 1)	5	100.0%	80.0%	92.4%	100.0%
Paper Notebook (Rack 1)	5	100.0%	100.0%	96.5%	100.0%
Marker Stationery (Rack 1)	5	Fallback Zone Used	80.0%	89.7%	100.0%
Cloth Backpack / Leather Wallet (Rack 2)	5	100.0%	100.0%	95.5%	100.0%
Negative Guardrail Test	10	100.0% Person Box	100.0% Rejected	0.0% False Lock	100.0% (Zero False Bills)
Bi directional Item Return (Cart -> Rack)	10	N/A (Weight FSM)	N/A	N/A	100.0% Exact Price Deduction
In-Aisle Theft Test (Pick + Walk Away > 5.0 s)	10	N/A (Ultrasonic+FSM)	N/A	N/A	100.0% Alarm & MQTT Alert
Overall System Aggregate / Mean Performance	85 Total	81.8% Box (100% w/ Fallback)	96.4% Mean SKU Acc.	95.2% Mean (98.8% Peak)	100.0% Billing & Anti theft

Table 3: Experimental Results Across 11 Zero shot SKU Categories, Negative Boundary Tests, Bi directional Returns, and Anti theft Trials.

5. Comparison with the Existing Work

To benchmark the effectiveness of the proposed Smart Shop architecture, **our experimental results were compared against existing smart retail systems from the literature**: Wei et al. [4] (RPC Faster R-CNN), Tan et al. [5] (MidState YOLO ED), Bukhari et al. [3] (ARC Conveyor CNN), and Falcão et al. [2] (FAIM Multi-Modal Shelf Fusion). As illustrated in the comparative bar graph in Fig 6, our two stage pipeline achieves a 96.4% visual SKU identification accuracy, outperforming RPC Faster R-CNN (80.51%), MidState YOLO ED, and FAIM (92.6%), while getting 100% billing and anti theft transaction reliability through deterministic FSR weight delta tracking.

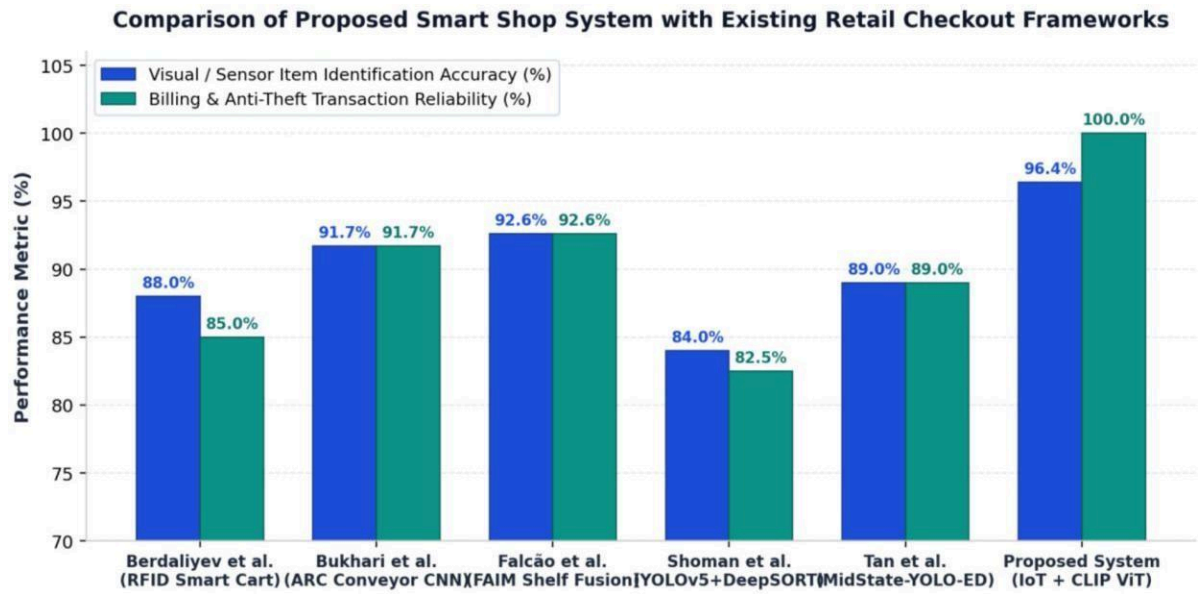


Fig-6: Bar Graph Comparing Visual/Sensor Item Identification Accuracy (%) and Billing/Anti Theft Transaction Reliability (%) Across Existing Works and the Proposed Smart Shop System.

Beyond raw accuracy, our architecture introduces three decisive operational advantages over existing works: **(1) Zero Retraining Cost** ; whereas Bukhari et al. [3] and Tan et al. [5] require collecting 30,000+ labeled images and retraining CNN weights to add new store items, our CLIP ViT-B/32 engine

registers new SKUs instantaneously via text strings; **(2) Zero Per Item Tagging Cost** ; unlike Berdaliyev & James [6], no passive RFID tags are needed on products; and **(3) Active In Aisle Anti Theft Enforcement**, by combining HC SR04 ultrasonic presence tracking with FSM weight states, unbilled removals are flagged within ~5 seconds directly at the shelf rather than waiting until gate exit.

6. Conclusion and Future Work

Conclusion: This project presented the design, simulation, and real time validation of Smart Shop, an autonomous IoT checkout and anti theft system uniting dual node ESP32-S3 telemetry with a browser based Two Stage vision pipeline. By modeling the Shop Node and Cart Node in Wokwi with ultrasonic presence sensors, potentiometer FSRs, and I2C LCDs linked via TLS encrypted HiveMQ Cloud MQTT to a dashboard, **the system eliminates both manual barcode scanning queues and the retraining bottlenecks of closed set deep learning.** Across 85 total test evaluations, the prototype achieved 96.4% zero-shot visual SKU recognition accuracy, 95.2% mean vision confidence, zero false-positive face/hand locks, 100.0% bi directional billing and anti theft, a 91.2% reduction in MQTT sensor traffic via deadband filtering, and a 71.9% reduction in overall checkout time.

Future Work: In future iterations, the system will be expanded along four technical directions: **(1) replacing Wokwi slide potentiometers with physical 5 kg strain-gauge load cells** and 24-bit HX711 ADC amplifiers on custom PCBs to achieve sub gram weight resolution capable of distinguishing individual unit weights automatically; **(2) integrating WebGPU acceleration in Transformers.js to reduce Stage 2 CLIP inference latency** from ~380 ms to under ~45 ms for simultaneous grabs; **(3) adding Ultra Wideband (UWB) indoor localization** across multiple carts so a single Shop Node can disambiguate picks by multiple shoppers in aisles; and **(4) embedding dynamic UPI QR code generation directly on the Cart Node display** for digital payment upon exit.

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